



EXAM PAPERS PRACTICE

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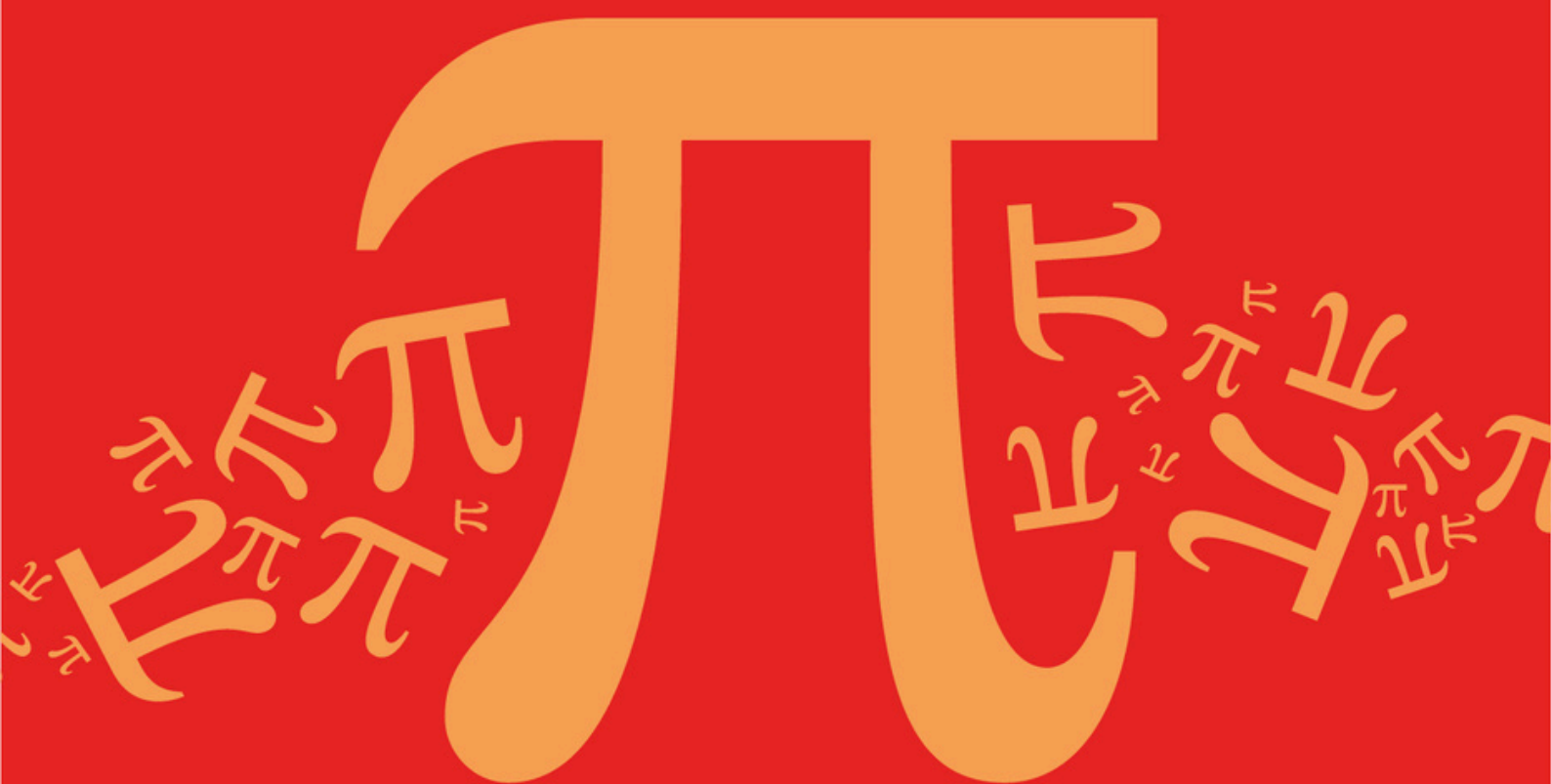
Practice questions created by actual examiners and assessment experts

Detailed mark scheme

Suitable for all boards

Designed to test your ability and thoroughly prepare you

AQA A Level Further Mathematics (7367)



Topic

D: Further Algebra and Functions

Sub topic

D16: Transformations of Graphs

Mark Scheme

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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Mark schemes

Q1.

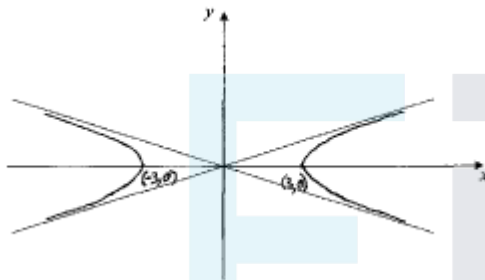
(a) $y = \pm \frac{1}{3}x$

ACF Need both

B1

1

(b)



2-branch curve with branches in correct regions above and below x-axis

B1

Curve approaching asymptotes

B1

*Coords $(\pm 3, 0)$, as **only** points of intersection with coordinate axes, indicated. Condone -3 and +3 marked on x-axis at points of intersection as $(\pm 3, 0)$ indicated.*

B1

3

(c) (i) $\frac{(x+3)^2}{9} - y^2 = 1$

Replacing x by either $x + 3$ or $x - 3$

M1

ACF

A1

2

(ii) $\frac{(x+3)^2}{9} - x^2 = 1$

Substitution into c's (c)(i) eqn of $y = x$ to eliminate y or of $x = y$ to eliminate x

M1

$$x^2 + 6x + 9 = 9(x^2 + 1)$$

Correct expansion of $(x \pm 3)^2$ equated to $9(x^2 + 1)$ OE ft;
[OE in y]

A1F

$$8x^2 - 6x = 0 \quad (8x^2 = 6x)$$

Ft on error $(x - 3)$ for $(x + 3)$ in (c)(i) which gives

$$8x^2 + 6x = 0 \quad (8x^2 = -6x) \text{ [OE in y]}$$

A1F

Points are (0,0), $(\frac{3}{4}, \frac{3}{4})$

Both. ACF

A1

4

- (d) Adding 3 to c's (c)(ii) **two** x-coords keeping
y-coordinates unchanged.

M1

Points are (3,0), $(3\frac{3}{4}, \frac{3}{4})$

Ft on c's (c)(ii) coordinates for the **two** points
If not deduced then M0A0

A1F

2

EXAM PAPERS PRACTICE [12]

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Q2.

- (a) $(\pm\sqrt{5}, 0)$, $(0, \pm 2)$

If not B2, award B1 if either at least two of these 4 correct
pts or if ' $x = \pm\sqrt{5}$ and $y = \pm 2$ '

B2,1

2

(b) $\frac{(x-p)^2}{5} + \frac{y^2}{4} = 1$

Replacing x by either $x + p$ or $x - p$ and keeping y
unchanged or as $y \pm 0$

M1

ACF

A1

2

(c) $\frac{(x-p)^2}{5} + \frac{(x+4)^2}{4} = 1$

Substitution into c's (b) eqn of $y = x + 4$ to eliminate y

M1

$$4(x - p)^2 + 5(x + 4)^2 = 4 \times 5$$

$$4(x^2 - 2px + p^2) + 5(x^2 + 8x + 16) = 20$$

$$4x^2 - 8px + 4p^2 + 5x^2 + 40x + 80 = 20$$

*Denominators 5 and 4 cleared in a correct manner
and at least either a correct expansion of $(x \pm p)^2$ or
a correct expansion of $(x + 4)^2$*

m1

$$9x^2 - (8p - 40)x + 4p^2 + 60 = 0$$

CSO No errors in any line of working.

*AG. Must see brackets correctly removed and all terms
involving x , p correctly rearranged to same side before
the printed answer is stated. Must have ' $= 0$ ' although
brackets around $4p^2 + 60$ may be omitted*

A1

3

(d) Discriminant is $(8p - 40)^2 - 4(9)(4p^2 + 60)$

OE Must be isolated, not just within the quadratic formula

B1

For tangency

$$(8p - 40)^2 - 4(9)(4p^2 + 60) = 0$$

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OE Equating c 's discriminant to zero before obtaining
any values for p*

M1

$$p^2 + 8p + 7 = 0$$

ACF with like terms collected

A1

$$\{(p + 1)(p + 7) = 0 \Rightarrow \} p = -1, p = -7(*)$$

Correct values $-1, -7$ for p

B1

$$p = -1: 9x^2 + 48x + 64 (= 0)$$

$$p = -7: 9x^2 + 96x + 256 (= 0)$$

*Substitutes at least one of c 's **two** values for p either
into the given quadratic in (c) OE or into $\frac{8p - 40}{18}$*

M1

$$p = -1: 9x^2 + 48x + 64 (= 0) \Rightarrow x = -\frac{8}{3}$$

$x = -\frac{8}{3}$ OE as only root from the quadratic or from
 $\frac{8p-40}{18}$. Apply FIW if (*) is B0

A1

$$p = -7: 9x^2 + 96x + 256 (= 0) \Rightarrow x = -\frac{16}{3}$$

$x = -\frac{16}{3}$ OE as only root from the quadratic or from
 $\frac{8p-40}{18}$ Apply FIW if. (*) is B0

A1

$$x = -\frac{8}{3}, y = \frac{4}{3}, x = -\frac{16}{3}, y = -\frac{4}{3}$$

$$\left(-\frac{8}{3}, \frac{4}{3}\right) \quad \left(-\frac{16}{3}, -\frac{4}{3}\right)$$

CSO Previous 7 marks must have been awarded and coordinates of both points need to be correct and exact but accept in either format

A1

EXAM PAPERS PRACTICE ⁸ [15]

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Q3.

- (a) Sketch of ellipse
centred at origin

M1

Correct relationship to circle

A1

Coords $(\pm 2\sqrt{2}, 0), (0, \pm \sqrt{2})$

Accept $\sqrt{8}$ for $2\sqrt{2}$;

B1 for any 2 of $x = \pm 2\sqrt{2}, y = \pm \sqrt{2}$

allow B1 if all correct except for use of decimals (at least one DP)

B2,1

4

- (b) (i) Replacing x by $\frac{x}{2}$

or by $2x$

M1

$$E \text{ is } \left(\frac{x}{2}\right)^2 + y^2 = 2$$

OE

A1

2

(ii) Tangent is $\frac{x}{2} + y = 2$

M1 for complete valid method

M1A1

2

[8]

Q4.

(a) (i) Elimination to give $x = \frac{1}{8}x^2$
OE

M1

A is (8, 8)

NMS 2/2

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A1

2

(ii) Equation of Q is $x = \frac{1}{8}y^2$

OE; condone $y = \sqrt{8x}$

B1

1

(iii) Points of contact are images in $y = x$

E1

1

(b) (i) Eliminating y to give $-x + c = \frac{1}{8}x^2$

M1

(ie $x^2 + 8x - 8c = 0$)

Distinct roots if $\Delta > 0$

stated or implied

E1

$$\Delta = 64 + 32c, \text{ so } c > -2$$

convincingly shown (AG)

A1

3

(ii) For tangent $c = -2$, so $x^2 + 8x + 16 = 0$

OE

M1

... and $x = -4, y = 2$

A1

Reflection in $y = x$
or other complete method

M1

$x = 2, y = -4$

ft wrong answer for first point; allow NMS 2/2

A1F

4

EXAM PAPERS PRACTICE [11]

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Q5.

(a) Ellipse with centre of origin

Allow unequal scales on axes

B1

$(\pm \sqrt{3}, 0)$ and (0 ± 2) shown on diagram

Condone AWRT 1.7 for $\sqrt{3}$;
B1 for incomplete attempt

B2,1

3

(b) y replaced by $\frac{1}{2}y$

M1A0 for $2y$ instead of $\frac{1}{2}y$

M1A1

Equation is now
$$\frac{x^2}{3} + \frac{y^2}{16} = 1$$

A1

3

- (c) Attempt at completing the square

M1

$$4(x - 1)^2 + 3(y + 1)^2 \dots$$

A1A1

[Alt: replace x by $x - a$ and y by $y - b$

M1 if one replacement correct

(M1)

$$4x^2 - 8ax + 3y^2 - 6by \dots]$$

Condone errors in constant terms

(m1A1)

$$a = 1 \text{ and } b = -1$$

A1A1

5

[11]

Q6.

- (a) Asymptotes are $y = \pm \frac{1}{2}x$
OE; M1 for $y = \pm mx$

M1A1

2

- (b) $x = 4$ substituted into equation

M1

$$y^2 = 3 \text{ so } y = \pm \sqrt{3}$$

Allow NMS

A1

2

- (c) (i) y -coords are $2 \pm \sqrt{3}$
ft wrong answer to (b)

B1F

1

- (ii) Hyperbola is $\frac{x^2}{4} - (y - 2)^2 = 1$

M1A0 if $y + 2$ used

M1A1

Asymptotes are $y = 2 \pm \frac{1}{2}x$
ft wrong gradients in (a)

B1F

3

[8]

Q7.

- (a) Curve translated 7 in y direction

B1

... and 1 in negative x direction
or answer in vector form

B1

2

- (b) (i) Asymptotes $x = -1$ and $y = 7$

B1B1

2

- (ii) Intersections at $(0, 8)$...

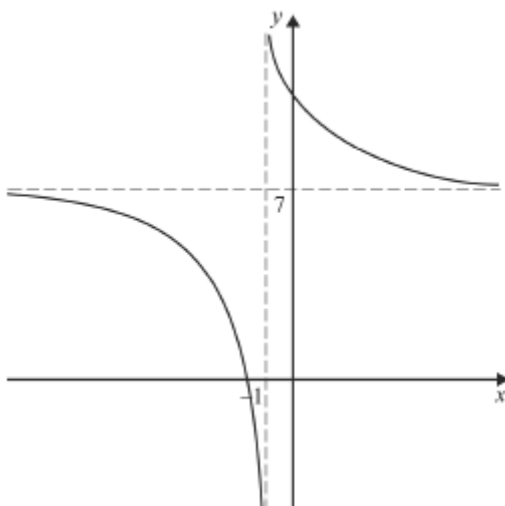
B1

... and $(-\frac{8}{7}, 0)$
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Allow AWRT -1.14 ; NMS $1/2$

M1A1

3

- (c)



At least one branch

of correct shape

B1

Complete graph

translation of $y = 1/x$

B1

All correct including asymptotes

in roughly correct positions

B1

3

[10]

Q8.

- (a) Intersections $(\pm\sqrt{2}, 0)$, $(0, \pm 1)$
Allow B1 for $(\sqrt{2}, 0)$, $(0, 1)$

B1B1

2

- (b) Equation is $\frac{(x-k)^2}{2} + y^2 = 1$
M1 if only one small error, eg $x + k$ for $x - k$

M1A1

2

- (c) Correct elimination of y

M1

Correct expansion of squares

M1

Correct removal of denominator

M1

Answer convincingly established
 AG

A1

4

- (d) Tgt $\Rightarrow 4(k+4)^2 - 12(k^2+6) = 0$

M1

.... $\Rightarrow k^2 - 4k + 1 = 0$

OE

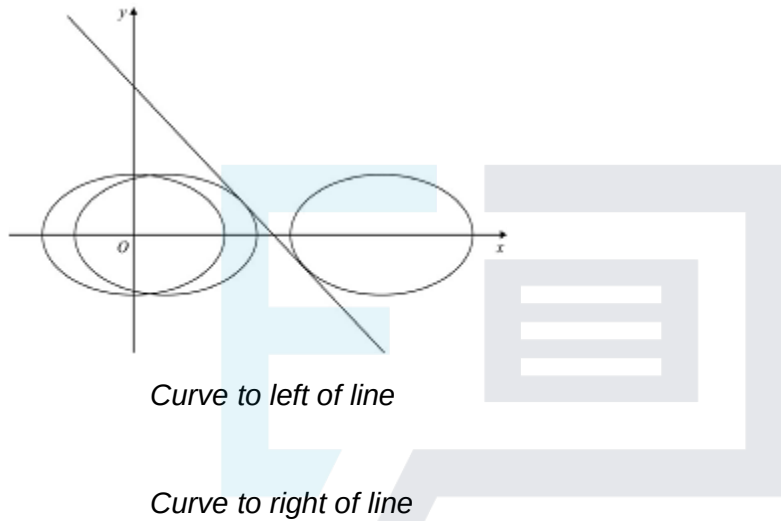
m1A1

$$\dots \Rightarrow k = 2 \pm \sqrt{3}$$

A1

4

(e)



B1

B2

3

Curves must touch the line in approx correct positions
 SC 1/3 if both curves are incomplete but touch the line correctly

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[15]

Q9.

(a) Good attempt at sketch

M1

Correct at origin

A1

2

(b) (i) y replaced by $y - 2$

B1

Equation is $(y - 2)^2 = 12x$
 ft $y + 2$ for $y - 2$

B1ft

2

(ii) Equation is $x^2 = 12y$

B1
1

(c) (i) $(x + c)^2 = x^2 + 2cx + c^2$

B1

... = $12x$

M1

Hence result
convincingly shown (AG)

A1
3

(ii) Tangent if $(2c - 12)^2 - 4c^2 = 0$

M1

ie if $-48c + 144 = 0$ so $c = 3$

A1
2

(iii) $x^2 - 6x + 9 = 0$

M1

$x = 3, y = 6$

A1
2

(iv) $c = 4 \Rightarrow$ discriminant = $-48 < 0$
OE

M1A1

So line does not intersect curve

A1
3

[15]

Q10.

(a) Stretch parallel to y axis ...

B1

... scale-factor $\frac{1}{2}$ parallel to y axis

B1
2

(b) $(x - 2)^2 - y^2 = 1$

M1A1

Translation in x direction ...

A1

... 2 units in positive x direction

A1

4

[6]

Q11.

(a) Correct shape

Coordinates $(\pm 3, 0), (0, \pm 2)$ shown

Allow labels on sketch

B1

B2,1

3

(b) Attempt to solve

$$\frac{1}{9} + \frac{y^2}{4} = 1$$

M1

At least one correct root

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Allow decimals; allow

$$\sqrt{\frac{32}{9}}$$

m1

$$y = \pm \frac{4}{3} \sqrt{2}$$

A1

3

(c) Eqn is $\frac{(x-1)^2}{9} + \frac{y^2}{4} = 1$

M1A0 for eg wrong sign

M1A1

2

[8]

Examiner reports

Q1.

This question, involving hyperbolas, was generally answered well although the final part was frequently not answered in the required manner.

In part (a) some gave the wrong equations $y = \pm 3x$ for the asymptotes and some others did not even give an equation.

Most candidates, in part (b), were able to show the main features of the curve in their sketches and to state the correct coordinates of the points of intersection with the axis although some other sketches were seen which bore little resemblance to the shape of a hyperbola.

As expected, the most common wrong answer for part (c)(i) was ' $\frac{(x-3)^2}{9} - y^2 = 1$ ' but this error was not overly penalised as some follow through was allowed in the later part. There were, however, many correct solutions to both parts (c)(i) and (c)(ii).

Generally only the more able candidates were able to **deduce** the correct coordinates in part (d). Candidates who started afresh, making no reference to their answers in part (c)(ii), were awarded no credit. Common wrong follow through answers were obtained by either subtracting 3 from their x-coordinates in part (c)(ii) or adding / subtracting 3 to / from their y-coordinates in part (c)(ii).

Q2.

The majority of students had a clear understanding of the methods required to answer this final question which was based on the equation and translation of an ellipse. In part (a) most students were able to find the values of x when $y = 0$ and the values of y when $x = 0$ but a significant minority of students either gave the coordinates incorrectly or did not attempt to express the values in coordinate form. In part (b), the majority of students gave

the correct equation of the translated ellipse but the incorrect answers $\frac{(x+p)^2}{5} + \frac{y^2}{4} = 1$

and $\frac{x^2 - p^2}{5} + \frac{y^2}{4} = 1$ were also seen. In part (c), the correct substitution of $y = x + 4$ was generally used and most indicated that they knew how to clear brackets and fractions but in attempting to reach the printed result, algebraic working was not always absolutely correct. In such questions, each line of the student's solution has to be correct and any corrections made must be made to all relevant lines of working. In part (d), the majority of students correctly equated their discriminant to zero although some inequalities were also seen. Those who used $(40 - p)$ as their value for 'b' were slightly more successful than those who used $-(p - 40)$. Those who failed to arrive at the correct quadratic equation in p, which they then tried to solve, often knew they had to substitute their value(s) into the given quadratic in part (c) but could go no further. A minority of students however, having obtained the correct equation $p^2 + 8p + 7 = 0$ then surprisingly gave the x-coordinates as -1 and -7. Although sign errors were sometimes seen in substituting -1 and -7 for p into the quadratic in part (c), the majority of students who obtained the correct quadratic equations in x usually went on to find the correct coordinates of the points of contact.

Q3.

In part (a) of this question, most candidates managed to draw an acceptable attempt at an

ellipse touching the given circle in the appropriate places. Occasionally the stretch would be applied parallel to the y -axis rather than the x -axis. The required coordinates were indicated with various levels of accuracy, sometimes appearing as integers, sometimes with minus signs omitted.

In part (b), the most successful candidates were often the ones who wrote the least — all that

was needed was to replace x by $\frac{x}{2}$ in each of the two given equations. Many candidates answered part (b)(i) concisely and correctly but then went into a variety of long methods to find the equation of the tan gent in part (b)(ii). Some used implicit differentiation and were usually successful. Others used chain-rule differentiation and usually made errors. Yet others embarked on a very complicated piece of work based on quadratic theory and using a general gradient m . This must have consumed a large amount of time and was almost invariably unsuccessful.

Q4.

In part (a)(i), some candidates found the short method of using the equation $y = x$ to find the coordinates of A . More commonly, the equation of the parabola Q was used in conjunction with that of P , and after some relatively heavy algebra the required coordinates were found correctly. Most candidates were able to write down the equation of Q in part (a)(ii), or indeed had already written it down in their working for the first part of the question. Only a handful of candidates gave an adequate answer in part (a)(iii). Many stated that the common tan gent had to be perpendicular to the mirror line, but for this to be entirely convincing it was necessary to mention the two points of contact of the line with the parabolas, which must be mirror images of each other.

Part (b)(i) gave candidates the opportunity to show their knowledge of quadratic theory, though in some cases they failed to keep to the rules for correct manipulation of inequalities. In part (b)(ii), some candidates immediately saw that c must be equal to -2 for the tangency case, while others virtually repeated their work done in the previous part, with equals signs, to establish the correct value of c . Once the coordinates of one point of contact had been found, it was possible to write down the other pair by using a reflection in $y = x$, but many candidates performed another calculation to find the second pair of coordinates.

Q5.

The sketches of the ellipse in part (a) of this question were mostly satisfactory, though some candidates failed to take square roots when working out the required coordinates.

Part (b) was one of those parts of questions where the absence of any explanation made it hard for the examiners to see what the candidates were trying to do. Some gave the answer in simplified form without any preliminary working. On this occasion the examiners condoned this poor examination technique where it looked as if the candidate had done the right thing. But there were many errors here: some modified the x term rather than the y term, some multiplied the y by 2 instead of dividing it by 2, and many failed to include this 2 in the squaring process when simplifying their equations.

Part (c) was found to be rather hard, but many candidates realised that completing the square was needed and at least made an effort, often failing to see the connection between their correct equations and the required numbers a and b . The alternative approach of applying the translation from the outset, using the letters a and b , was about

as popular as the completing the square method, but the algebra became too heavy for many candidates and the necessary comparing of terms did not take place.

Q6.

The oblique asymptotes of a hyperbola were not always known by the candidates, though many found the necessary general equations in the formula booklet and correctly applied them to this particular case. Part (b) was very well answered by almost all candidates, while part (c) usually provided some further marks, the most common error being to use $y + 2$ instead of $y - 2$ in the equations of the translated hyperbola and asymptotes.

Q7.

Many candidates scored well on this question but, for some, marks were lost in a variety of places.

In part (a) some candidates were careless in giving the two parts of the translation.

In part (b) (i) the horizontal asymptote was often found to be the x -axis, which usually caused difficulty in part (c). In part (b)(ii) a mark was sometimes lost by a failure to show some necessary working, in this case for the intersection of the curve with the x -axis.

In part (c) the sketches were sometimes wildly wrong, despite the information provided in

part (a) that the curve must be a translated version of the well-known hyperbola $y = \frac{1}{x}$. In many cases the curve shown was basically correct but did not appear to approach the asymptotes in a satisfactory way. Despite a generous interpretation of this on the part of the examiners, some candidates lost credit because of seriously faulty drawing.

Q8.

This final question presented a varied challenge, mostly focused on the algebraic skills needed to cope with the quadratic equation printed in part (c) of the question – deriving this equation from two other equations and then using the discriminant of the equation to find the cases where the ellipse would touch the line. Many candidates were well prepared for this type of question and scored heavily, though with occasional errors and omissions.

Part (e) of the question was often not attempted at all, or the attempts were totally incorrect, involving transformations other than translations parallel to the x -axis. In some cases only one of the two possible cases was illustrated, usually the one where the ellipse touched the straight line on the lower left side of the line, but an impressive minority of scripts ended with an accurate portrayal of both cases.

Q9.

Marks were often lost in part (a) by a sketch which showed only the upper part of the parabola, or by the appearance of a cusp at the origin instead of a smooth curve touching the y -axis.

Part (b)(i) was also answered incorrectly by many candidates, who inserted $+ 2$ or $- 2$ in inappropriate places. The transformation in part (b)(ii) was better known.

In part (c) many candidates sailed through all the parts in good style and picked up ten marks out of ten.

Part (c)(i) was a particularly straightforward test for almost all the candidates. In part (c)(ii) a common mistake was to give the discriminant as $(2c - 12)^2 - 4$, ignoring the c^2 which was the constant term of the quadratic in part (c)(i). Some candidates omitted the y -coordinate in part (c)(iii), or gave it as ± 6 . Part (c)(iv) provided most candidates with the opportunity to end on a high note with a brief but correct response.

Q10.

In order to gain full marks in this question it was necessary to use the correct standard technical words 'stretch' and 'translation'. In part (a) most candidates did indeed use the word 'stretch', but very rarely with the correct scale factor. The candidates were not always familiar with the process of completing the square, but when they did carry out this requirement correctly they usually went on to give the correct transformation in part (b).

Q11.

Part (a) was usually answered correctly, although some candidates seemed to spend more time on it than they would have had if they had been more familiar with the context. Weaker candidates struggled with the algebra in part (b), but the more confident ones missed the significance of the plurals in the wording of the question and found only one y -coordinate. Most candidates wrote down their answers in part (c) as required, often correctly but sometimes with an extraordinary variety of errors.